

LECTURE 1.2



Expectation and Conditional Means

ECONXXXXX Econometrics I

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PLACEHOLDER TERM

Where We Are

1. Foundations

1.1 Data and Variation

1.2 Conditional Means

1.3 Sampling Variation

2. Simple Regression

2.1 Fitting a Line

2.2 Goodness of Fit

2.3 Bias and Precision

2.4 Reporting an Estimate

3. Multiple Regression

3.1 Adding Controls

3.2 Functional Form

3.3 Joint Tests

4. Limits

4.1 Diagnosing Violations

4.2 Limits of Regression

Last Week's Lecture

Lecture 1.1 set the module's question and the vocabulary for describing data:

1 How This Module Works.

What the term covers, how it is assessed, and the standing question.

2 Populations and Samples.

The question is about a population; the data is a sample.

3 Describing One Variable.

Centre, spread and shape, and what each one hides.

4 Describing Two Variables.

Covariance, correlation, and what one number cannot show.

Lecture Plan

1 **Expectation.**

A property of a distribution, not of a sample.

2 **The Algebra of Expectations.**

The rules every derivation in this module uses.

3 **The Conditional Expectation Function.**

The best guess of earnings given schooling.

SECTION 1

Expectation

1. Expectation
2. The Algebra of Expectations
3. The Conditional Expectation Function

Defining the Expectation of a Random Variable

A random variable is a quantity the draw decides. Its expectation says where the distribution sits, before any sample:

$$E[y] = \sum_j y_j p_j \quad (1)$$

- y : the variable, here hourly earnings in euro
- y_j : which value, running over every value y can take
- p_j : one value that y can take
- p_j : the share of the population sitting at that value
- $E[y]$: the population average of y , here 11.52 euro

Notation for the Population Mean and the Sample Mean

From here every summary comes as a pair: one population, one sample.

The Population Mean Is Fixed, the Sample Mean Is Not

$E[y]$ is one fixed number, a property of the distribution. $\bar{y}$ is the sample average, 12.25 in our 120. Draw a different 120 and $\bar{y}$ is a different number.

The Population Mean Is Not Directly Observable

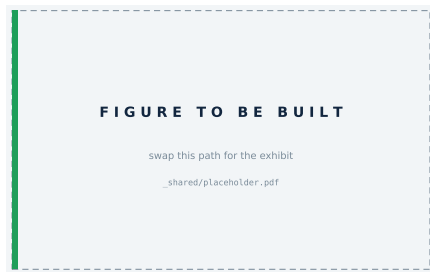


Figure 1: Population earnings in grey, the sample of 120 in colour, both means marked.

Only the Sample Mean Changes

Redraw the 120: the dashed line leaves 12.25, the solid stays at 11.52. $E[y]$ belongs to the population, not to the draw.

The Gap Between the Two Lines Has Its Own Distribution

Every claim this module makes is about the solid line. Every number we compute comes from the dashed one.

Lectures 1.3 and 2.3 Measure the Gap

The 0.73 between 12.25 and 11.52 is not a mistake to be removed. That gap changes with every sample drawn. A standard error puts a number on it.

SECTION 2

The Algebra of Expectations

1. Expectation
2. The Algebra of Expectations
3. The Conditional Expectation Function

The Three Rules of Expectation Algebra

Measuring that gap means averaging combinations of variables. Three rules do that without knowing the combination's distribution:

$$E[c] = c \qquad E[a + b y] = a + b E[y] \qquad E[y + z] = E[y] + E[z] \qquad (2)$$

- c : a fixed number, with no distribution of its own
- a, b : a shift and a scaling, applied identically to every draw
- z : a second random variable, related to y or not; the rule holds either way

A Constant Is the Same in Every Draw

An expectation averages across draws. A constant does not change from draw to draw, so averaging it returns it unchanged. A fixed multiplier factors out of the sum. On the right-hand side only y and z still carry an E , because only they vary.

Variance and Covariance Written as Expectations

Spread and co-movement are themselves averages, so the three rules reach them:

$$\text{Var}(y) = E[(y - E[y])^2] = E[y^2] - (E[y])^2 \quad (3)$$

$$\text{Cov}(y, x) = E[(y - E[y])(x - E[x])] = E[yx] - E[y]E[x] \quad (4)$$

- x : years of schooling
- $\text{Var}(y)$: the population variance, written σ_y^2 in Lecture 1.1
- $\text{Cov}(y, x)$: the population co-movement of the two
- $E[y^2]$: the average of the squares, not the square of the average

Why Expectations Do Not Pass Through a Product

An expectation passes through sums, constants and scalings, and not through a product. Independent means knowing one variable tells you nothing about the other. Last week the covariance of schooling and earnings in our 120 came out positive, so these two are not independent.

The Product Rule Fails by the Covariance

$E[yx] = E[y]E[x] + \text{Cov}(y, x)$. The two agree exactly when the covariance is zero. Independence gives that; the equality does not need it.

The Variance of a Sum Carries a Covariance Term

Adding two random variables does not add their variances:

$$\text{Var}(y + z) = \text{Var}(y) + \text{Var}(z) + 2 \text{Cov}(y, z)$$

(5)

Zero Covariance Drops the Term

The covariance term drops out when y and z are uncorrelated. Independence is stronger than this rule asks for.

Where Each Rule Is Used Later in the Module

- **Linearity:** cuts the Lecture 2.3 proof of unbiasedness to two lines from twenty.
- **The Variance of a Sum:** builds next week's standard error, and 2.3's.
- **Covariance Over Variance:** $\text{Cov}(y, x) / \text{Var}(x)$ is the slope Lecture 2.1 fits.

Each Rule Is the Sum Rearranged

The appendix writes the weighted sum, splits it, then takes the constants outside.

The Average of Schooling Times Earnings

Average schooling is 13.54 years and average earnings 11.52 euro.

Question

Is the average of (schooling $\times$ earnings) about 156?

SECTION 3

The Conditional Expectation Function

1. Expectation
2. The Algebra of Expectations
3. The Conditional Expectation Function

Defining the Conditional Expectation of Earnings Given Schooling

A covariance says these two vary together. It does not say what to expect at a given level of schooling. Split by schooling, then average inside each group:

$$E[y \mid x] \quad (6)$$

- $E[y \mid x]$: read “given”; the average of y among people with that x

In a sample it is the mean from Lecture 1.1, taken over a chosen group:

$$\bar{y}_{x=12} = \frac{1}{n_{12}} \sum_{i: x_i=12} y_i \quad (7)$$

- $\sum_{i: x_i=12}$: add over only the people with twelve years
- n_{12} : how many of our 120 have twelve years

Earnings Distributions at Three Levels of Schooling

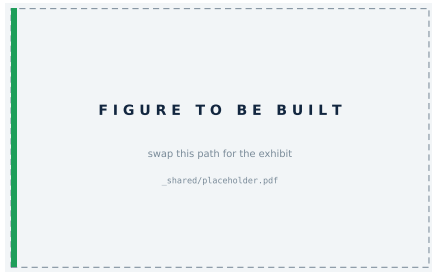


Figure 2: Earnings among people with ten, twelve and sixteen years of schooling.

The Three Distributions Overlap

Many people with sixteen years of schooling earn less than the average at ten. The generator sets the spread around each mean at 4.50.

The Conditional Expectation Function in This Population

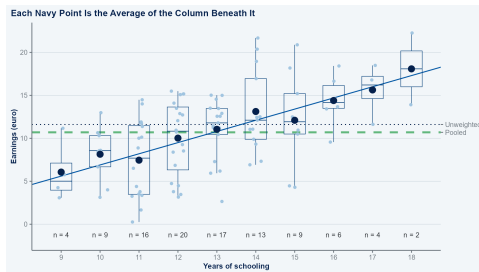


Figure 3: One hundred people drawn for this picture; each navy point is that column's average.

The Counts Are the Weights

The blue line is the CEF written into the generator, $E[y | x] = 9.50 + 1.30 (x - 12)$. Pooling all one hundred gives 10.70. Averaging the ten column means without their counts gives 11.61, weighting two people at eighteen years like twenty at twelve.

The Sense in Which the Conditional Mean Is Best

No rule built on schooling beats the conditional mean on squared error:

$$E[y | x] = \arg \min_g E[(y - g(x))^2] \quad (8)$$

- $g(x)$: any function of x used to guess y
- $\arg \min_g$: the g that makes what follows as small as it goes

Both Expected Squares Must Be Finite

Both $E[y^2]$ and $E[g(x)^2]$ must be finite. Without them there is no expected squared error to compare.

Why the Minimum Sits at the Conditional Mean

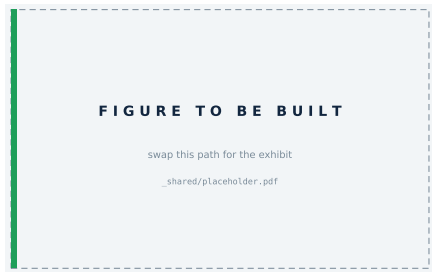


Figure 4: Expected squared error at twelve years of schooling, against the guess made.

The Minimum Is the Conditional Variance

The curve is lowest at a guess of 9.50, at a height of 4.50^2 . That height is the conditional variance, and no guess goes below it.

What the Fitted Line in Lecture 2.1 Approximates

Without the CEF, a fitted line is the easiest curve to draw. Angrist & Pischke (2015) build the regression chapters of *Mastering Metrics* on the CEF.

The Line Approximates the CEF

Lecture 2.1 fits a line by least squares to approximate the CEF. This population's CEF is straight; real ones need not be. The CEF exists in the population before anyone fits anything.

Description Versus Causation in the Conditional Mean

Describing who earns what is not the same as saying what schooling does.

A Description, Not an Effect

$E[y \mid x]$ describes the population and says nothing about changing x . “People with sixteen years of schooling earn 14.70 on average” is a description. “Four more years raises earnings by 5.20” is a different claim. Nothing today supports it.

What Would Have To Be True?

Today's number is the conditional mean of earnings at each level of schooling.

Question

What would have to be true for this number to mean what we just said it means? And what is the plausible story where it isn't?

Application

The CEF explorer cuts schooling into one-year bins. Each bin gets one point, at its mean of earnings.

Question

With our 120 people, will the bin means look like a line?

The lab repeats the exercise on all 200,000.

Summary: Expectation, Its Algebra, and the CEF

Today was three ideas, each doing one job:

1 **Expectation.**

A property of the distribution, not of the data you hold.

2 **The Algebra of Expectations.**

Three rules carry every derivation, and none of them covers a product.

3 **The Conditional Expectation Function.**

The best guess at earnings, and a description rather than an effect.

Before Lecture 1.3

1 Play:

The CEF explorer, widening the bin from one year to four.

2 Apply:

Problem set 1, covering 1.1 and 1.2, with both appendix derivations.

3 Review:

Wooldridge (2020), Math Refresher B-4e and B-4f, on conditional expectation.

Next Week

Lecture 1.3 computes an expectation from 120 people instead of a population:

- **Estimators:** rules that turn 120 numbers into one guess at $E[y]$.
- **Sampling Variation:** how far the guess lands from $E[y]$ across samples.
- **Intervals and Tests:** two claims about the procedure, not about the number.

Thank you

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BACKUP

Appendix

Derivation: The Algebra of Expectations

Write each expectation as its weighted sum, split it, take the constants outside.

Three Rules, the Variance Form Follows

Three rules follow: a constant, a linear function, and a sum. The variance form $E[y^2] - (E[y])^2$ is a consequence of them, not a fourth rule.

Derivation: The Conditional Mean Minimises Expected Squared Error

Add and subtract $E[y \mid x]$ inside the square, then expand. The cross term has expectation zero.

The Cross Term Has Mean Zero

Assume $E[y^2]$ and $E[g(x)^2]$ are finite. What remains is the conditional variance plus a squared term, zero only at $g(x) = E[y \mid x]$.

Notes and Tools

- **Binning in R:** `cut()` and `tapply()`, which is the whole binned-means lab.
- **The Lecture in One Line:** `mean(y[x == 12])` against `mean(y)`.
- **This Week's Script:** the CEF explorer, and the code that draws its exhibits.

References

Angrist, J. D., & Pischke, J.-S. (2015). *Mastering 'metrics: The path from cause to effect*. Princeton University Press.

Wooldridge, J. M. (2020). *Introductory econometrics: A modern approach* (7th ed.). Cengage Learning.